



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2019 Year 12 Course Trial Examination

Friday, 16 August 2019

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order behind this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided

SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: **# BOOKLETS USED:**

Class: (please ✓)

☐ 12MAT.1 – Mr Lam

☐ 12MAT.2 – Mrs Bhamra

☐ 12MAT.3 – Mrs Gan

☐ 12MAT.4 – Ms Park

☐ 12MAT.5 – Mr Sekaran

Marker's use only

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	

Section I

10 marks

Attempt Questions 1 to 10

Allow approximately 10 minutes for this section

Mark your answers on the answer grid provided.

Questions

Marks

1. Which expression is equal to $\int \sin^2 2x \, dx$? 1

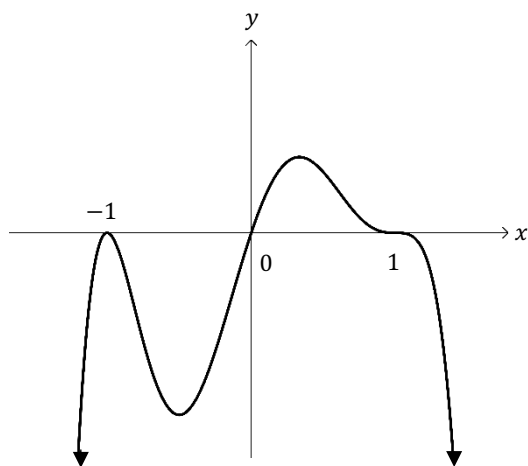
(A) $\frac{1}{8}(4x + \sin 4x) + c$

(B) $\frac{1}{8}(4x - \sin 4x) + c$

(C) $-\frac{\cos^3 2x}{6} + c$

(D) $\frac{\sin^3 2x}{6} + c$

2. Which of the following could be the equation of the graph given below? 1



(A) $y = -\frac{x}{2}(1-x)^3(1-x)^2$

(B) $y = -\frac{x}{2}(1-x)^2(x+1)^3$

(C) $y = 3x(1-x)^3(x+1)^2$

(D) $y = 4x(1-x)^2(x-1)^3$

3. $x = 0.5$ is the first approximation to the positive root of the function $f(x) = xe^x - 1$. 1

Which of the following is the second approximation using Newton's method, correct to two decimal places?

(A) $x = 0.56$

(B) $x = 0.57$

(C) $x = 0.58$

(D) $x = 0.59$

4. Which of the following is equal to $\lim_{x \rightarrow 0} \left(\frac{\sin 4x}{2x \cos 2x} \right)$? 1

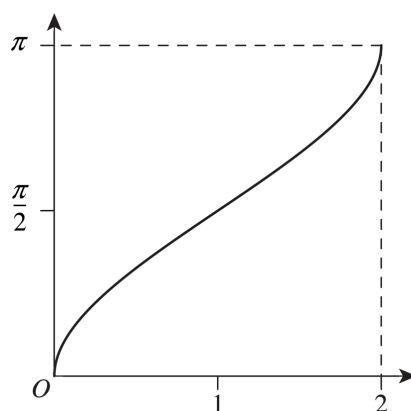
- (A) $\frac{1}{2}$
(B) 0
(C) 4
(D) 2

5. The point P divides the interval from $A(2, -5)$ to $B(-4, 1)$, internally in the ratio 1:2. 1

What are the coordinates of P ?

- (A) $(-\frac{14}{3}, \frac{5}{3})$
(B) $(-\frac{14}{3}, -3)$
(C) $(0, -3)$
(D) $(-8, 11)$

6. Which function best describes the graph below? 1



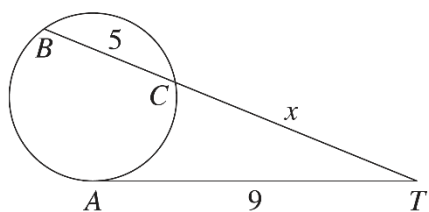
- (A) $y = \cos^{-1} x$
(B) $y = 1 - \cos^{-1} x$
(C) $y = \cos^{-1}(x - 1)$
(D) $y = \cos^{-1}(1 - x)$

7. Which of the following is equal to $\frac{d}{dx}(\cos^{-1} \frac{x}{4})$? 1

- (A) $-\frac{1}{\sqrt{4-x^2}}$
 (B) $-\frac{1}{2\sqrt{4-x}}$
 (C) $-\frac{4}{\sqrt{16-x^2}}$
 (D) $-\frac{1}{\sqrt{16-x^2}}$

8. The line AT is the tangent to the circle at A and the line BT is a secant meeting the circle at B and C . $AT = 9$, $BC = 5$ and $CT = x$. 1

Which of the following equations is correct?



- (A) $x^2 + 5x - 81 = 0$
 (B) $x^2 + 5x + 81 = 0$
 (C) $x^2 - 5x - 81 = 0$
 (D) $x^2 + 5x - 9 = 0$

9. If $y = \cos(\ln x)$, which is equal to $\frac{d^2y}{dx^2}$? 1

- (A) $\frac{\cos(\ln x) + \sin(\ln x)}{x^2}$
 (B) $\frac{x^2 \cos(\ln x) + \sin(\ln x)}{x^2}$
 (C) $\frac{\sin(\ln x) - x^2 \cos(\ln x)}{x^2}$
 (D) $\frac{\sin(\ln x) - \cos(\ln x)}{x^2}$

10. Which inequality has the same solution as $|x + 3| + |x - 4| = 7$? 1

- (A) $|2x - 1| \geq 7$
 (B) $x^2 - x - 12 \leq 0$
 (C) $\frac{7}{3-x} \geq 1$
 (D) $\frac{1}{x-4} - \frac{1}{x+3} \leq 0$

Section II

60 marks

Attempt Questions 11 to 14

Allow approximately 1 hour and 50 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Marks

- (a) Find all the asymptotes of $y = \frac{5x}{x^2+x-12}$. **2**
- (b) Solve $\frac{1}{2x} \geq -\frac{1}{x-2}$. **4**
- (c) The roots of $x^2 - 2x - 1 = 0$ are $\tan \alpha$ and $\tan \beta$. **3**
Given that α and β are acute, find the exact value of $\alpha + \beta$.
- (d) Find the size of the acute angle between the curves $f(x) = (x + 2)^2 + 4$ and $g(x) = x^2 - 4$ at their point of intersection. Give your answer to the nearest degree. **3**
- (e) The region enclosed by the graph of $y = \frac{1}{\sqrt{1+9x^2}}$, the line $x = 0$, the line $x = \frac{1}{3}$ and the x -axis **3**
is rotated about the x -axis to form a solid of revolution.

Find the exact volume of the solid formed.

Question 12 (15 marks)

Use a SEPARATE Writing Booklet.

- (a) A student tosses two regular dice with faces numbered 1 to 6. He records the maximum of the two uppermost faces as a score.

- i. Find the probability that he records the score 1 in a single throw of the two dice. 1
- ii. Find the probability that he records the score 6 in a single throw of the two dice. 1

- (b) Consider the polynomial $P(x) = x^3 + bx^2 + cx - 18$. 3

Given that $(x + 2)$ is a factor of $P(x)$ and -8 is the remainder when $P(x)$ is divided by $(x + 1)$, find the values of b and c .

- (c) Evaluate, by using the substitution $u^2 = 1 + x$: 3

$$\int_0^3 \frac{x(x+2)}{\sqrt{1+x}} dx$$

- (d) i. Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2
- iii. Hence or otherwise, find the general solution of the equation $\cos x - \sqrt{3} \sin x = \sqrt{2}$. 2

- (e) The rate at which a body cools in air is assumed to be proportional to the difference between its temperature T and the constant room temperature P of the surrounding air. This can be expressed as:

$$T = P + Ae^{kt}$$

where t is time in hours and k is a constant.

A cup of tea cools from 85°C to 25°C in 90 minutes. The air temperature P in the room is 20°C .

- i. Show that $k = \frac{2}{3} \log_e \frac{1}{13}$. 2
- ii. Find the temperature of the cup of tea after a further 30 minutes. 1
Give your answer correct to 2 decimal places.

Question 13 (15 marks)

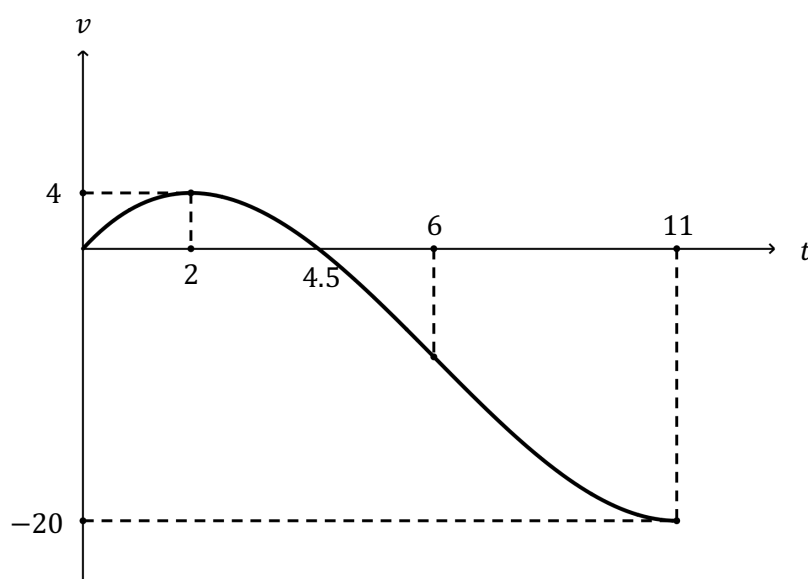
Use a SEPARATE Writing Booklet.

- (a) By letting
- $t = \tan \theta$
- , prove that:

2

$$\frac{\tan 2\theta + \cot \theta}{\tan 2\theta - \tan \theta} = \cot^2 x$$

- (b) The velocity,
- $v \text{ ms}^{-1}$
- , of a particle moving along a straight line is shown in the graph below.

Initially, the particle is at rest at the origin and returns to the origin at $t = 7$.*Diagram not drawn to scale.*

- i. When is the speed of the particle the greatest? 1
 - ii. How many times does the particle change direction? 1
 - iii. When is the acceleration of the particle the greatest after it starts moving? 1
 - iv. Sketch the graph of the displacement function $x(t)$ for $0 \leq t \leq 7$, showing all important features. 2
- (c) Consider the function $f(x) = \sqrt{2x - 1} + 1$.
- i. Find the equation of the inverse function $f^{-1}(x)$, stating the domain. 2
 - ii. Find the x -coordinates of the points where the graphs $y = f(x)$ and $y = f^{-1}(x)$ intersect. 2

- (d) Liquid flows into a container made in the shape of a hollow, inverted right circular cone with radius 16 cm and height 24 cm, as shown in the diagram.

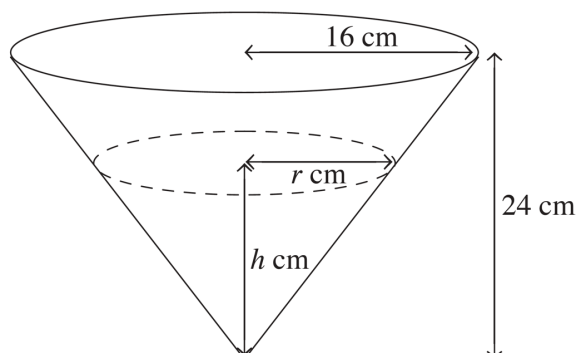


Diagram not drawn to scale.

At time t seconds, the height of the liquid is h cm, the surface of the liquid has radius r cm and the volume of the liquid is V cm³.

- i. Show that the $V = \frac{4\pi h^3}{27}$. 2
- ii. Liquid flows into the container at a rate of $8 \text{ cm}^3 \text{ s}^{-1}$. 2

Show that $\frac{dh}{dt} = \frac{1}{2\pi}$ when $h = 6$.

Question 14 (15 marks)

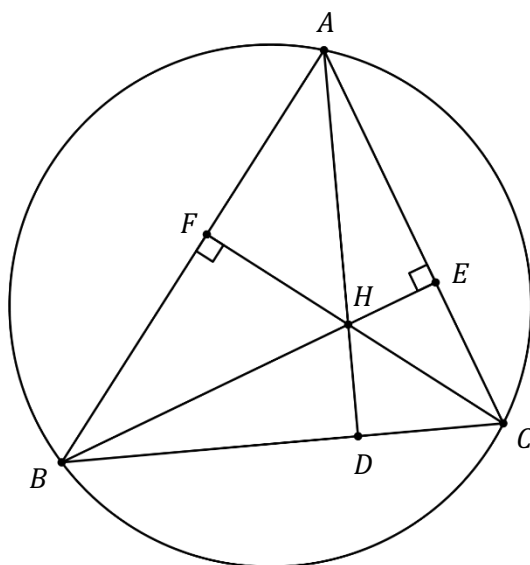
Use a SEPARATE Writing Booklet.

- (a) Use mathematical induction to prove that, for every positive integer
- n
- :

3

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4}n^2(n+1)^2$$

- (b) Consider the following acute-angled triangle
- ABC
- .

*Diagram not drawn to scale.*

Points E and F lie on AC and AB respectively, such that BE is perpendicular to AC and CF is perpendicular to AB . The line segments BE and CF meet at H .

The line segment AH produced meets BC at D .

Copy or trace the diagram into your writing booklet.

- i. Give a reason for why $CEFB$ is a cyclic quadrilateral. 1
- ii. By considering cyclic quadrilaterals, show that $\angle CAD = \angle EFH = \angle EBC$, giving full reasons. 3
- iii. Hence, prove that AD is perpendicular to BC . 2

- (c) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola with equation $x^2 = 4ay$, where $p \neq 0, q \neq 0$ and $p \neq q$. The parabola has a focus $S(0, a)$.

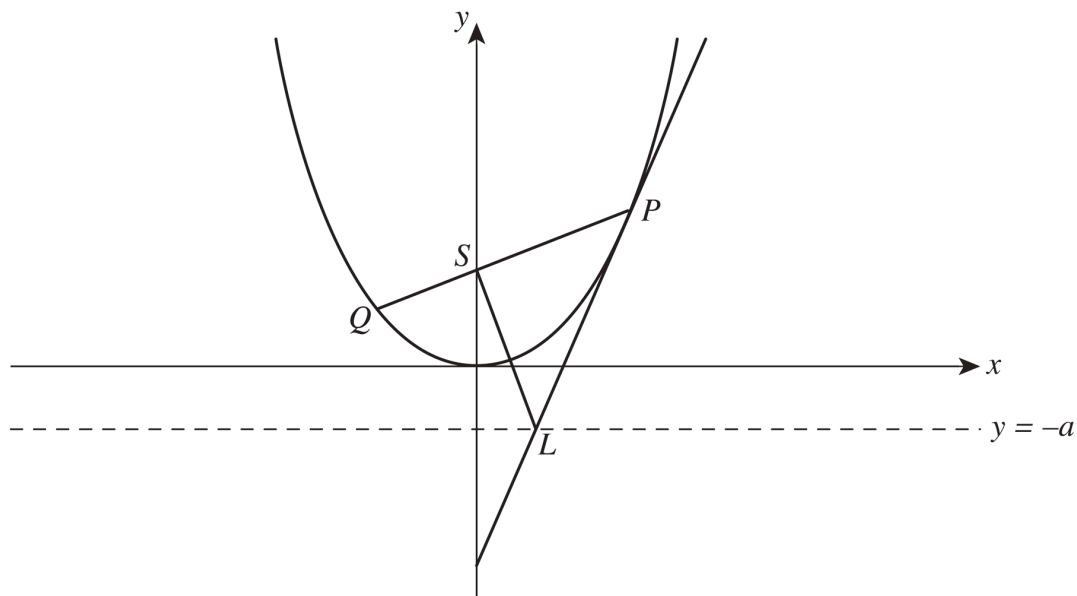


Diagram not drawn to scale.

Let L be the intersection of the tangent at P and the directrix, and PQ is a focal chord.

- i. Show that $pq = -1$. 1
- ii. Explain why $PS = a(p^2 + 1)$. 1
- iii. Show that $PS \times QS = LS^2$. 4

End of examination.

Q1) $\int \sin^2 2x \, dx$

$$= \int \frac{1}{2} (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} (x - \frac{1}{4} \sin 4x) + C$$

$$= \frac{1}{8} (4x - \sin 4x) + C \quad (B)$$

Q2) (C)

Q3) $f(x) = xe^x - 1$

$$f'(x) = xe^x + e^x$$

$$= e^x(x+1)$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.5 - \frac{f(0.5)}{f'(0.5)}$$

$$= 0.5 - \frac{0.5e^{0.5} - 1}{e^{0.5}(0.5+1)}$$

$$= 0.5710 \dots$$

$$\div 0.57 \quad (B)$$

Q4) $\frac{\sin 4x}{2x \cos 2x} = \frac{2 \sin 2x \cos 2x}{2x \cos 2x}$

$$= \frac{\sin 2x}{x}$$

$$= 2 \cdot \frac{\sin 2x}{2x}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin 4x}{2x \cos 2x} = 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$$

$$= 2 \cdot 1$$

$$= 2 \quad (D)$$

Q5) $A(2, -5) \quad B(-4, 1)$

$$1:2$$

$$P = \left(\frac{1x - 4 + 2x - 5}{1+2}, \frac{1x + 2x - 5}{1+2} \right)$$

$$= (0, -3) \quad (C)$$

Q6) The graph shows $y = \cos^{-1}x$ reflected in the y -axis shifted right by 1. (D)

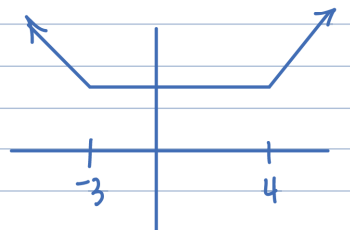
$$\begin{aligned} \text{Q7) } \frac{d}{dx} \cos^{-1} \frac{1}{4}x \\ &= - \frac{1}{\sqrt{1 - (\frac{x}{4})^2}} \cdot \frac{1}{4} \\ &= - \frac{1}{4 \sqrt{\frac{16-x^2}{16}}} \\ &= - \frac{1}{\sqrt{16-x^2}} \end{aligned} \quad (D)$$

$$\begin{aligned} \text{Q8) } AT^2 &= CT \times BT \\ q^2 &= x \cdot (x+5) \\ 0 &= x^2 + 5x - 81 \end{aligned} \quad (A)$$

$$\begin{aligned} \text{Q9) } y &= \cos(\ln x) \\ y' &= -\frac{1}{x} \sin(\ln x) \\ &= -x^{-1} \sin(\ln x) \\ y'' &= 1x^{-2} \sin(\ln x) + -x^{-1} \cdot \frac{1}{x} \cos(\ln x) \\ &= \frac{\sin(\ln x) - \cos(\ln x)}{x^2} \end{aligned}$$

(D)

$$\text{Q10) } |x+3| + |x-4| = \begin{cases} -2x+1, & x < -3 \\ 7, & -3 \leq x \leq 4 \\ 2x-1, & x > 4 \end{cases}$$



$$\begin{aligned} x^2 - x - 12 &\leq 0 \\ (x+3)(x-4) &\leq 0 \\ -3 &\leq x \leq 4 \end{aligned}$$

(B)

Q11) a) $y = \frac{5x}{x^2 + x - 12}$
 $= \frac{5x}{(x-3)(x+4)}$

asymptotes: $x = 3, -4$ ✓
 $y = 0$ ✓

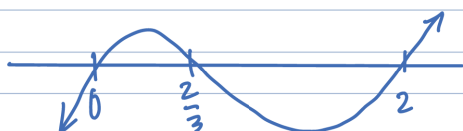
b) $\frac{1}{2x} \geq -\frac{1}{x-2}, x \neq 0, 2$

$x(x-2)^2 \geq -2x^2(x-2)$ ✓ multiply by $2x^2(x-2)^2$

$x(x-2)^2 + 2x^2(x-2) \geq 0$

$x(x-2)[x-2 + 2x] \geq 0$

$x(x-2)(3x-2) \geq 0$ ✓



$\therefore 0 < x \leq \frac{2}{3}, x \geq 2$ ✓
 ✓ (excl. $x=0, 2$)

c) $x^2 - 2x - 1 = 0$

$\tan \alpha + \tan \beta = \frac{-(-2)}{1}$
 $= 2$ }

$\tan \alpha \cdot \tan \beta = \frac{-1}{1}$
 $= -1$ }

$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$
 $= \frac{2}{1 - (-1)}$
 $= 1$ }

$\alpha + \beta = \frac{\pi}{4}$ ✓

$$d) f(x) = g(x)$$

$$(x+2)^2 + 4 = x^2 - 4$$

$$x^2 + 4x + 4 + 4 = x^2 - 4$$

$$4x = -12$$

$$x = -3 \quad \checkmark$$

$$f'(x) = 2(x+2)$$

$$m_1 = 2(-3+2)$$

$$= -2$$

$$g'(x) = 2x$$

$$m_2 = 2(-3)$$

$$= -6$$

$$\tan \theta = \left| \frac{-2 - (-6)}{1 + (-2)(-6)} \right|$$

$$= \frac{4}{13}$$

$$\theta = \tan^{-1}\left(\frac{4}{13}\right)$$

$$= 17.10 \dots$$

$$\doteq 17^\circ \quad \checkmark$$

$$e) V = \pi \int y^2 dx$$

$$= \pi \int_0^{\frac{1}{3}} \frac{1}{1+9x^2} dx \quad \checkmark$$

$$= \pi \int_0^{\frac{1}{3}} \frac{1}{1+(3x)^2} dx$$

$$= \pi \left[\frac{1}{3} \tan^{-1}(3x) \right]_0^{\frac{1}{3}} \quad \checkmark$$

$$= \frac{\pi}{3} \left(\tan^{-1}\left(3 \cdot \frac{1}{3}\right) - \tan^{-1}(0) \right)$$

$$= \frac{\pi}{3} \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi^2}{12} \text{ units}^3 \quad \checkmark$$

Q(2) a)

	1	2	3	4	5	6
1	0	.	.	.	.	.
2	.	0	.	.	.	.
3	.	0	.	.	.	.
4	.	.	.	.	.	.
5	.	.	.	.	.	.
6	.	.	.	.	.	.

i) $P(1,1) = \frac{1}{36}$ ✓

ii) $P(6,6) + P(6, \text{other}) + P(\text{other}, 6) = \frac{11}{36}$ ✓

b) $P(x) = x^3 + bx^2 + cx - 18$

$$P(-2) = (-2)^3 + b(-2)^2 + c(-2) - 18 = 0$$

$$-8 + 4b - 2c - 18 = 0$$

$$2b - c = 13 \quad \text{--- ①}$$

$$P(-1) = (-1)^3 + b(-1)^2 + c(-1) - 18 = -8$$

$$-1 + b - c - 18 = -8$$

$$b - c = 11 \quad \text{--- ②}$$

$$\textcircled{1} - \textcircled{2} : b = 2$$

$$\hookrightarrow 2 - c = 11$$

$$c = -9$$

$$c) \quad u^2 = 1+x$$

$$x = u^2 - 1$$

$$dx = 2u \, du$$

$$x=3 \rightarrow u=2$$

$$x=0 \rightarrow u=1$$

$$\int_0^3 \frac{x(x+2)}{\sqrt{1+x}} \, dx = \int_1^2 \frac{(u^2-1)(u^2+1)}{\sqrt{u^2}} \cancel{2u} \, du \quad \checkmark$$

$$= 2 \int_1^2 u^4 - 1 \, du$$

$$= 2 \left[\frac{u^5}{5} - u \right]_1^2$$

$$= 2 \left(\frac{2^5}{5} - 2 - \left(\frac{1^5}{5} - 1 \right) \right)$$

$$= 2 \cdot \frac{26}{5}$$

$$= \frac{52}{5} \quad \checkmark$$

$$d) \quad i) \quad \cos x - \sqrt{3} \sin x = R \cos(x + \alpha) \\ = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\left. \begin{aligned} 1 &= R \cos \alpha \\ \sqrt{3} &= R \sin \alpha \end{aligned} \right\} \quad \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3} \quad \checkmark$$

$$R^2 = 1 + 3$$

$$R = 2 \quad \checkmark$$

$$\therefore \cos x - \sqrt{3} \sin x = 2 \cos\left(x + \frac{\pi}{3}\right)$$

$$ii) \quad \cos x - \sqrt{3} \sin x = \sqrt{2}$$

$$2 \cos\left(x + \frac{\pi}{3}\right) = \sqrt{2}$$

$$\cos\left(x + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}} \quad \checkmark$$

$$x + \frac{\pi}{3} = 2\pi n \pm \frac{\pi}{4}, \quad n \in \mathbb{Z}$$

$$x = 2\pi n \pm \frac{\pi}{4} - \frac{\pi}{3} \quad \checkmark$$

$$e) \quad i) \quad T = P + Ae^{kt}$$

$$t=0 \rightarrow 85 = 20 + Ae^0$$

$$A = 65 \quad \checkmark$$

$$t=1.5 \rightarrow 25 = 20 + 65e^{k(1.5)}$$

$$\frac{1}{13} = e^{1.5k}$$

$$\log_e \frac{1}{13} = 1.5k$$

$$k = \frac{2}{3} \log_e \frac{1}{13} \quad \checkmark$$

$$ii) \quad t=2 \rightarrow T = 20 + 65e^{k(2)}$$

$$= 22.1264 \dots$$

$$\approx 22.13^\circ \quad \checkmark$$

Q(3) a) $LHS = \frac{\tan 2\theta + \cot \theta}{\tan 2\theta - \tan \theta}$

$$= \left(\frac{2t}{1-t^2} + \frac{1}{t} \right) \div \left(\frac{2t}{1-t^2} - t \right), \quad \tan \theta = t$$

$$= \frac{2t^2 + 1 - t^2}{t(1-t^2)} \div \frac{2t - t(1-t^2)}{1-t^2} \quad \checkmark$$

$$= \frac{t^2 + 1}{t(1-t^2)} \times \frac{1-t^2}{t+t^3}$$

$$= \frac{t^2 + 1}{t} \times \frac{1}{t(1+t^2)} \quad \checkmark$$

$$= \frac{1}{t^2}$$

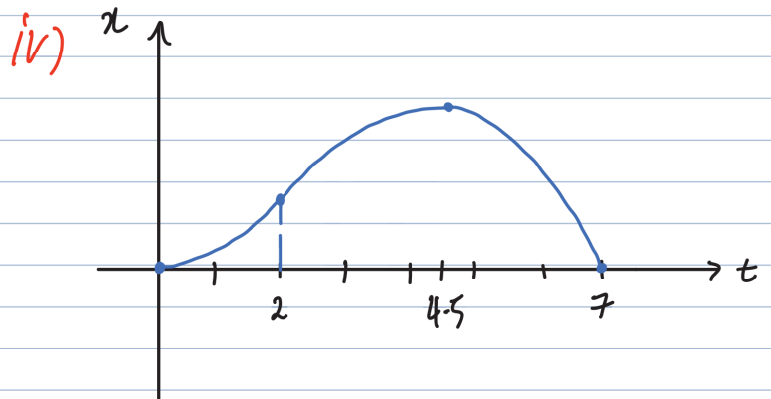
$$= \cot 2\theta$$

$$= RHS$$

b) i) $t = 11$ (20 m/s)

ii) once (at $t = 4.5$)

iii) $t = 6$



✓ shape

✓ { turning pts $x=0, 4.5$
and $x=7$
pt of inflexion $x=2$

c) i) $y = \sqrt{2x-1} + 1$

$$x = \sqrt{2y-1} + 1$$

$$(x-1)^2 = 2y-1$$

$$y = \frac{1}{2} ((x-1)^2 + 1), \quad x \geq 1 \quad \checkmark$$

ii) $\sqrt{2x-1} + 1 = x$

$$2x-1 = (x-1)^2$$

$$= x^2 - 2x + 1$$

$$0 = x^2 - 4x + 2$$

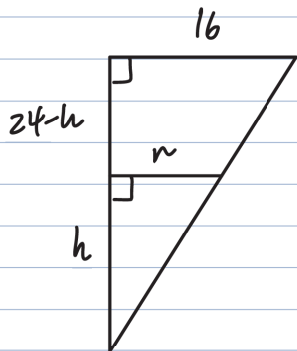
$$x = \frac{4 \pm \sqrt{4^2 - 4(1)(2)}}{2(1)}$$

$$= \frac{4 \pm \sqrt{8}}{2}$$

$$= 2 \pm \sqrt{2}$$

$$= 2 + \sqrt{2} \quad (x > 1) \quad \checkmark$$

d) i)



$$\frac{r}{h} = \frac{16}{24}$$

$$r = \frac{2h}{3} \quad \checkmark$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \left(\frac{2h}{3} \right)^2 \cdot h$$

$$= \frac{4\pi}{27} h^3 \quad \checkmark$$

ii) $\frac{dV}{dt} = 8$

$$\frac{dV}{dh} = \frac{4\pi}{9} h^2 \quad \checkmark$$

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$8 = \frac{4\pi}{9} h^2 \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{18}{\pi h^2}$$

$$h=6 \rightarrow \frac{dh}{dt} = \frac{18}{36\pi} = \frac{1}{2\pi} \quad \checkmark$$

Q14) a) $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4} n^2 (n+1)^2$

prove true for $n=1$.

$$\text{LHS} = 1^3$$

$$\text{RHS} = \frac{1}{4} (1)^2 (1+1)^2$$

$$= 1$$

$$= \text{LHS}$$

$\therefore$ The result is true for $n=1$ ✓

Assume true for $n=k$.

$$\text{i.e. } 1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4} k^2 (k+1)^2$$

prove true for $n=k+1$.

$$\text{i.e. } 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{1}{4} (k+1)^2 (k+2)^2$$

$$\text{LHS} = 1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3$$

$$= \frac{1}{4} k^2 (k+1)^2 + (k+1)^3$$

$$= \frac{1}{4} (k+1)^2 [k^2 + 4(k+1)]$$

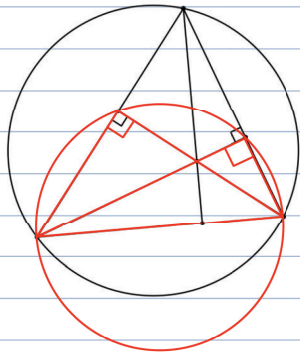
$$= \frac{1}{4} (k+1)^2 (k+2)^2$$

$$= \text{RHS}$$

$\therefore$ The result is true for $n=k+1$ ✓

Hence the statement is true for all positive integers n by mathematical induction. ✓

b) i)

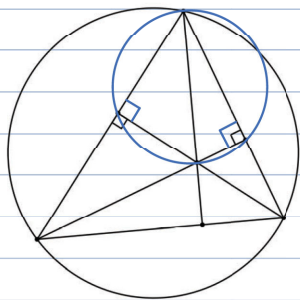


$$\begin{aligned}\angle CEB &= 180^\circ - \angle AEH \quad (\text{Ls on a straight line}) \\ &= 90^\circ \\ &= \angle CFB \quad (\text{given})\end{aligned}$$

$\therefore$ CEFB is a cyclic quad. ✓

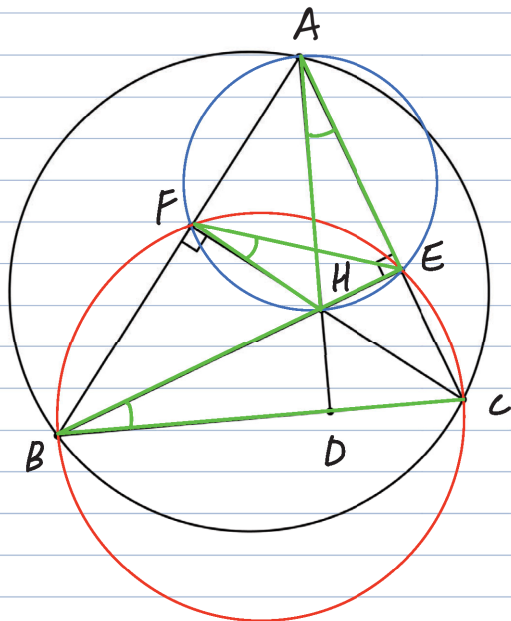
(interval BC subtends equal Ls on the same side so CEFB are concyclic)

ii)



$$\begin{aligned}\angle AFH &= 180^\circ - \angle BFC \quad (\text{Ls on a straight line}) \\ &= 90^\circ \\ &= \angle AEH \quad (\text{given})\end{aligned}$$

$\therefore$ EAFH is a cyclic quadrilateral ✓
(opposite Ls of a cyclic quad. are supplementary)



$$\angle CAD = \angle EFH \quad (\text{Ls in the same segment})$$

and

$$\angle EFH = \angle EBC \quad (\text{similarly}) \quad \checkmark$$

$$\therefore \angle CAD = \angle EFH = \angle EBC$$

iii) $\angle ADC = 180^\circ - \angle CAD - \angle DCA$ ($\angle$ sum of $\triangle ADC$)

$$= 180^\circ - \angle CAD - (90^\circ - \angle EBC) \quad \checkmark$$

$$= 90^\circ - \angle CAD + \angle EBC \quad \checkmark \quad (\angle CAD = \angle EBC \text{ proved in part ii})$$

$$= 90^\circ$$

$$\therefore AD \perp BC$$

$$\begin{aligned}
 \text{c) i) } m_{PQ} &= \frac{ap^2 - aq^2}{2ap - 2aq} \\
 &= \frac{(p+q)(p-q)}{2(p-q)} \\
 &= \frac{p+q}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{chord } PQ : y - \cancel{ap^2} &= \frac{p+q}{2} (x - 2ap) \\
 &= \frac{p+q}{2} x - ap(\cancel{p+q})
 \end{aligned}$$

$$y = \frac{p+q}{2} x - apq$$

$$\begin{aligned}
 (0, a) \rightarrow \text{chord } PQ : a &= 0 - apq \\
 pq &= -1
 \end{aligned}$$

Alternatively,

$$\begin{aligned}
 m_{PS} &= \frac{ap^2 - a}{2ap - 0} \\
 &= \frac{p^2 - 1}{2p}
 \end{aligned}$$

$$\begin{aligned}
 m_{SQ} &= \frac{aq^2 - a}{2aq - 0} \\
 &= \frac{q^2 - 1}{2q}
 \end{aligned}$$

$$m_{PS} = m_{SQ}$$

$$\frac{p^2 - 1}{2p} = \frac{q^2 - 1}{2q}$$

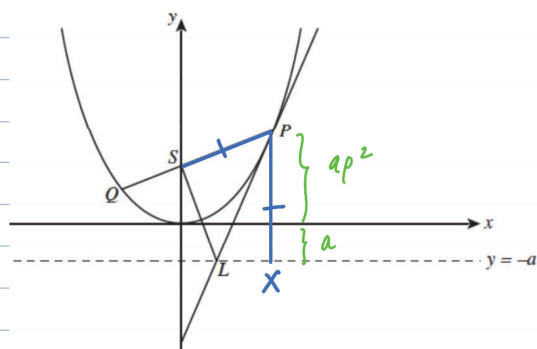
$$q(p^2 - 1) = p(q^2 - 1)$$

$$\begin{aligned}
 p - q &= pq^2 - qp^2 \\
 &= pq(q - p)
 \end{aligned}$$

$$pq = -1$$

or some other valid method

ii)



$$\begin{aligned} PS &= PX \text{ (locus definition of parabola)} \\ &= ap^2 + a \\ &= a(p^2 + 1) \end{aligned}$$

iii) tangent p : $y = px - ap^2$

directrix : $y = -a$

$$-a = px - ap^2$$

$$px = ap^2 - a$$

$$x = ap - \frac{a}{p}$$

$$\therefore L \left(ap - \frac{a}{p}, -a \right) \quad \checkmark$$

$$LS^2 = (2a)^2 + \left(ap - \frac{a}{p} \right)^2$$

$$= a^2 \left(4 + \left(p - \frac{1}{p} \right)^2 \right) \quad \checkmark$$

$$= a^2 \left(4 + p^2 - 2 + \frac{1}{p^2} \right)$$

$$= a^2 \left(p + \frac{1}{p} \right)^2$$

$$PS \times QS = a(p^2 + 1) \cdot a \left(\frac{1}{p^2} + 1 \right) \quad \checkmark$$

$$= a^2 (p^2 + 1) \left(\frac{1}{p^2} + 1 \right)$$

$$= a^2 (p^2 + 1) \left(\frac{1}{p^2} + 1 \right)$$

$$= a^2 (p^2 + 1) \left(\frac{1}{p^2} + 1 \right)$$

$$= a^2 \left(1 + p^2 + \frac{1}{p^2} + 1 \right)$$

$$= a^2 \left(p + \frac{1}{p} \right)^2 \quad \checkmark$$

$$= LS^2$$